

Section: Electricity

Topic: Current, PD, Power, Resistance

Question 2 – Potential Divider with Thermistor

(a) Calculate the total resistance of the circuit

✓ 3 marks

The **voltmeter** reads **4.6 V** across the **4.0 kΩ** resistor.

The **supply voltage** is **6.0 V**, so this is a **potential divider**.

Let R_T be the total resistance.

$$\frac{V_{4.0}}{V_{total}} = \frac{R_{4.0}}{R_T} \Rightarrow \frac{4.6}{6.0} = \frac{4.0}{R_T} \Rightarrow R_T = \frac{4.0 \times 6.0}{4.6} = 5.22 \text{ k}\Omega$$

✓ **Answer: 5.22 kΩ**

(b) Effect of replacing the 2.0 kΩ resistor with a thermistor

✓ 2 marks

As the **thermistor** **warms up**, its **resistance decreases**.

This causes:

- A **decrease in voltage** across the lower leg of the divider
- So a **greater voltage drop** across the **4.0 kΩ** resistor

✓ **Answer:**

As the thermistor warms, the voltmeter reading **increases**
because more of the supply voltage is dropped across the 4.0 kΩ resistor.

Revision Tips

- A **potential divider** splits voltage in proportion to resistance:

$$V = \frac{R_{\text{target}}}{R_{\text{total}}} \cdot V_{\text{total}}$$

- **Thermistors** have resistance that decreases with temperature
 - In a voltage divider: decreasing one resistance increases the voltage across the other
-